

Oracle AI Agent Studio: Deep Dive

1. Introduction to Oracle AI Agent Studio

- **Vision:** To move Generative AI beyond chatbots and integrate it as a **true co-worker** within enterprises. Imagine every employee having a personal AI co-pilot that understands business data, assists with tasks, and simplifies complex processes.
- **What it is:** AI Agent Studio acts as a **command center** to create, manage, and deploy this new digital workforce, directly **baked into Fusion Applications**. It represents "Enterprise AI, Built in. Not Bolted on".
- **Dual Approach:**
 - **Embedded AI:** Hundreds of pre-built agents and templates are **included with Fusion Apps** and ready for immediate use. Minor modifications are possible in the Studio.
 - **Custom AI:** Allows users to **build new agents from scratch** or **tweak existing ones** to perfectly fit unique business needs, offering highly tailored capabilities.

2. What are AI Agents?

AI Agents are **autonomous digital workers powered by LLMs**. They are designed to:

- **Interpret user goals** and decide how to execute them.
- **Create plans, take actions,** and **call tools/APIs** as needed.
- **Connect to enterprise systems** for business context.
- Deliver **task automation** and possess agency.

3. Key AI Agent Concepts & Components

- **Agents / Worker Agents:**
 - Assigned **specific sub-tasks** within a workflow.
 - Execute specific domains of work as directed by a supervisor or detailed instructions.
 - Can run concurrently and use tools to perform tasks.
 - Equipped with specific functionalities like accessing databases, interacting with APIs, or offering suggestions.
 - **Ideally handle 1 to 5 tools** for optimal performance, and no more than 10.

- **Supervisor / Orchestrator Agent:**
 - **Coordinates and decides which worker agent to call** to perform a specific task.
 - Oversees, manages, or evaluates the workflow of other agents.
 - Their prompts focus on **decision-making, assignment distribution, monitoring**, and quality control.
- **Agent Teams / Workflows:**
 - A structured sequence of steps or actions an AI Agent follows to accomplish a business task or answer a query.
 - Best for tasks that are **complex, modular, or require parallel processing**.
 - Typically consist of **3-5 agents**; if more than 5 are needed, it's recommended to split into separate workflows for clarity.
 - Offer advantages like **specialization, parallelization, modularity, and scalability** for complex projects.
- **Tools:**
 - **Connect agents to backend systems** and functionalities, such as external APIs or databases.
 - Enhance agent capabilities by allowing them to **fetch or manipulate data** and perform actions.
- **Topics:**
 - Refers to a **specific subject or area of conversation** an AI agent is designed to handle.
 - The **core content or focus** of interactions.
 - Provides **detailed instructions** for specific conversation scenarios.
 - Example: "Explain Oracle Cloud pricing to a new customer".
- **System Prompts:**
 - An instruction, rule set, or guiding context that **defines the agent's behavior, identity, constraints, or operational boundaries**.
 - Determines **how the AI responds** and what policies or personality it should follow.
 - Example: "You are a helpful customer support agent. Always greet the customer, use polite language and never share confidential information".

4. When to Use Single vs. Multi-Agent Teams

- **Single Agent:**
 - **Best for:** Simple, self-contained tasks with clear objectives and minimal dependencies.

- **Advantages:** Simplicity (easier to design, maintain), Speed (no coordination needed), Resource Efficiency.
- **Limitations:** Scalability for large, complex problems; potential lack of specialized skills.
- **Multi-Agents (Supervisor/Orchestrator):**
 - **Best for:** Tasks that are complex, modular, or require domain specialization and parallel processing.
 - **Advantages:** Specialization (distinct expertise for subtasks), Parallelization (concurrent subtasks), Modularity (flexible workflow design), Scalability for larger projects.

5. Practical Applications: Oracle's Pre-built Agents

Oracle provides a broad library of agents across various business functions:

- **Human Capital Management (HCM):**
 - **Benefits Analyst:** Assists employees with inquiries on benefits enrollments and coverage.
 - **Payslip Agent:** Clarifies details on payslips, deductions, and compensation.
 - **Talent Advisor (My Team):** Helps managers evaluate employee performance and potential.
- **Supply Chain & Manufacturing (SCM):**
 - **Supply Disruption Mitigator:** Proactively suggests solutions for supply chain problems.
 - **Maintenance Advisor:** Allows technicians to access relevant information from maintenance manuals.
 - **Outbound Compliance Advisor:** Ensures compliance with safety standards and goods routing.
- **Customer Experience (CX) - Sales & Service:**
 - **Service Triage Agent:** Categorizes and evaluates incoming service requests for faster processing.
 - **Account Advisor Agent:** Helps sales representatives gather insights about an account using CRM/ERP data and documents.
- **Enterprise Resource Planning (ERP) & Enterprise Performance Management (EPM):**
 - **Expenses Agent:** Can create expense reports from receipts and credit card charges.

- **Project Proposal Generation:** Automates the drafting of comprehensive project proposals.

6. Key Tools within AI Agent Studio

- **Business Object Tool:** Connects to Fusion Applications to **retrieve and perform CRUD operations** (Create, Read, Update, Delete) on core business data like Opportunities, Service Requests, Payslips, or Products.
- **Document Tool (RAG):** Enables **semantic search** on unstructured documents (e.g., policy documents, FAQs) to ground agent responses with citations.
- **Email Tool:** Allows agents to **write and send emails** based on predefined alert codes or templates, for notifications or alerts.
- **Deep Link Tool:** Generates **navigation links** to specific Fusion application pages with context, directing users to relevant forms or workflows.
- **External REST Tool:** Connects to **any external web service** with a public REST interface, supporting various HTTP methods and authentication.
- **Calculator Tool:** Performs **numerical computations** and business calculations.
- **User Session Tool:** Makes the **context of the end-user** within Fusion available to agents.
- **Human-in-the-Loop (HITL):** Allows for **human review and approval** steps at any point in the process for oversight, compliance, and validation.

7. Security & Governance

Oracle has built robust security and governance controls into the platform:

- **Defense-in-Depth Guardrails:** Protect data from threats like **prompt injection**, toxic language, and harmful imagery.
- **Zero Data Retention:** For external model providers.
- **Role-Based Access Control:** Ensures users only have access to build and manage agents for the parts of the business they are responsible for. This involves setting up custom admin roles and connecting them to specific permission groups (e.g., HCM, SCM).
- **AI Governance Office (AIIGO):** Oracle provides resources and best practices for organizations to set up their own AI Innovation & Governance Office to realize AI value while ensuring compliance with corporate policies and local regulations.
- **TRAP (Token Relay And Proxy):** A feature that allows secure invocation of third-party REST services, handling protected access.

8. Testing, Evaluation, and Optimization (METRO)

AI Agent Studio provides comprehensive tools for monitoring, evaluating, tracing, and reporting (METRO) to ensure high-performing, accurate, and reliable AI Agent solutions.

- **Automated Testing:** Generate responses for benchmark datasets and **compute correctness** as the primary metric.
- **LLM as a Judge (LLMaaJ):** Leverages advanced AI to assess AI quality by comparing agent output with expected answers, providing a correctness score and explanation.
- **Key Metrics:** Track **Accuracy (Quality)**, **Latency (Performance)**, and **Token Usage (Cost/Agent efficiency)**.
- **Tracing:** Offers a **step-by-step execution timeline** for multi-step agent executions, showing sequence, duration, and status, along with inputs, outputs, and contextual data.

9. Roadmap & Pricing

- **Roadmap Evolution:** The journey evolves from simple low-code solutions to sophisticated **supervisor-worker agents** and complex **multi-step workflows** that automate entire business processes. Future enhancements include new triggers, automation, and logic nodes for workflow agents.
- **Predictable Pricing:**
 - **Embedded AI:** Included at **no additional cost** with your existing Fusion subscription.
 - **OCI-hosted LLMs** (e.g., Llama, Cohere): **No charge with unlimited, unmetered use.**
 - **OpenAI LLM:** A **generous allocation of over 100 million free tokens per month** is included to help customers get started. Additional tokens are included with custom agent subscriptions.
 - **Custom AI Agents:** Priced with **predictable monthly fees per user or per employee**, avoiding consumption-based models. There is no cost to develop, with subscriptions only required when publishing to users.

10. Conclusion

Oracle AI Agent Studio empowers organizations to **design, deploy, and manage an intelligent digital workforce** that is natively integrated, secure, and highly customizable. It

provides the tools and framework to build AI agents that understand business context, reason through problems, and take actionable steps, transforming enterprise productivity.

11. Comparative Landscape: Oracle vs. Other AI Agent Platforms

Feature / Aspect	Oracle AI Agent Studio	AWS Bedrock Agents	Google Vertex AI Agent Builder	Salesforce AI Agent Builder
Positioning	Native to Oracle Fusion Applications — “Built in, not bolted on.”	General-purpose AI agent orchestration in AWS ecosystem.	AI agent creation integrated with Google Cloud + Vertex AI.	Focused on Salesforce CRM ecosystem and customer-facing workflows.
Integration	Deeply embedded into Fusion (ERP, HCM, SCM, CX, EPM). Includes pre-built agents for these domains.	Connects to AWS services (Lambda, API Gateway, Step Functions, RDS, DynamoDB, etc.). External connectors available.	Tightly coupled with Google Cloud APIs (Dialogflow CX, BigQuery, GCS). Designed for multi-modal use cases.	Natively integrated with Salesforce CRM, Service Cloud, Einstein 1 Platform.
Customization	Low-code Studio: create new agents or tweak pre-built templates. Agents can use Oracle Business Object Tool, Document Tool (RAG), REST, etc.	Developers design workflows with AWS-native tools and Bedrock models. Agents can chain reasoning steps, call APIs, invoke Lambda.	Offers agent orchestration with Dialogflow-style flows, tool invocation, and data grounding via Vertex AI Search.	Low-code “Builder” to create copilots/agents tailored to sales, service, and marketing flows.

Tools & Connectors	Prebuilt tools: Business Object CRUD, Document RAG, Email, REST, Deep Link into Fusion, Human-in-the-Loop.	Tools & APIs accessible via AWS ecosystem, external API calls, custom Lambda functions.	Tool integrations for APIs, knowledge grounding, and enterprise data connectors (e.g., search, BigQuery).	Connects directly to Salesforce data objects, workflows, and AppExchange ecosystem.
Security & Governance	Zero data retention (for external models), RBAC, AI Governance Office (AIIGO), TRAP proxy for secure external calls.	Security inherits AWS Identity & Access Management (IAM), VPC isolation, and data residency controls.	Security governed by GCP IAM, Vertex AI data governance, and model/data isolation.	Security governed by Salesforce's platform policies, data residency, and governance guardrails.
Testing & Monitoring	METRO framework: tracing, evaluation, automated testing, LLM-as-a-Judge (LLMaJ), benchmark scoring.	Monitoring through CloudWatch, logging, and Bedrock evaluation tools (in preview).	Evaluation with Vertex AI Experiments, prompt tracing, and continuous monitoring.	Monitoring via Salesforce dashboards, usage metrics, and trust guardrails.
Pricing Model	Predictable: Fusion-embedded AI free; OCI LLMs unlimited; OpenAI tokens allocated free (>100M/mo). Custom agents: flat subscription per user/employee (not	Pay-as-you-go for model inference and underlying AWS services. No bundled free tokens.	Pay-as-you-go based on Vertex AI usage (tokens, compute, storage).	Licensing bundled into Salesforce editions; AI Builder credits allocated per tenant.

	consumption-based).			
Best Fit	Enterprises running Oracle Fusion who want domain-specific AI copilots with deep process integration.	Cloud-native builders who want maximum flexibility with AWS services and infra.	Google Cloud customers focusing on multi-modal AI (text, image, video) and data-centric use cases.	Organizations centered on Salesforce CRM looking for customer support and sales copilots.

12. Comparative Landscape: Oracle vs. AI/R Agents Builder

Feature	Oracle Fusion AI Agent Studio	Air's AI Agents Builder
Primary Goal/Purpose	To create, manage, and roll out a new digital workforce for the enterprise, enabling task automation and agency. Aims to interpret user goals, create plans, take actions, and call tools/APIs within enterprise applications.	To develop sophisticated AI agents rapidly, aiming to overcome work overload, fragmented software systems, and innovation roadblocks by solving business challenges with Agentic AI.
Integration/Platform	Fusion-native , built directly into Oracle Fusion Applications. Offers seamless integration with existing application experiences, natively accessing Fusion Business Objects, APIs, and rules. Can connect to external data sources and agent frameworks.	Designed for scalability to run agents across any environment . (The source does not specify a native platform integration like Fusion Applications).
Target User	Developed with a low/no-code Studio Launch approach, targeting both administrative Line of Business (LOB) users and implicitly, developers.	Explicitly made for developers and non-developers , emphasizing that it requires no technical background to navigate.

Ease of Use/Interface	Described as a "command center" for managing the digital workforce, providing an intuitive interface for configuring and refining agent teams.	Features a user-friendly drag-and-drop interface .
Agent Types/Architecture	Supports Single Agents for self-contained tasks, and Multi-Agents structured as Supervisor/Orchestrator patterns. Also includes Agent Teams for complex, modular, or parallel processing, and Workflow "Sequential" Agents . Can form Hierarchical Agents .	Focuses on enabling users to create custom AI agents . While it mentions "Agentic AI," it does not explicitly detail specific agent architectures like Supervisor/Worker patterns in the provided text.
Tool/API Connectivity	Connects agents to backend systems via a wide range of tools: Business Object Tool (for Fusion data/CRUD operations), Document (RAG) Tool , Email Tool , Deep Link Tool , Calculator Tool , External REST Tool (for any web service), and User Session Tool .	Connects to over 100 APIs .
LLM Support	Supports OCI-hosted LLMs such as Llama and Cohere. Also supports OpenAI GPT 4.1mini . Offers the flexibility for users to Bring-Your-Own LLM (BYO LLM) .	Connects to the most popular language models in the market . (Specific models are not detailed in the provided information).
Pre-built Templates/Components	Provides hundreds of embedded AI features and agents as well as pre-defined, Oracle-built AI Agent templates designed for quick setup. Offers a catalog of pre-built templates for fast deployment.	The provided source does not explicitly mention pre-built templates or components beyond the ability to create custom components.
Customization Capabilities	Allows for advanced customizations and extensions to Oracle-built templates. Users can build AI agent workflows from scratch . Supports flexible	Offers flexibility to create custom components .

	workflow design through modularity.	
Security & Governance	Emphasizes enterprise-grade security with controls built into the platform from inception. Includes robust guardrails against prompt injection, toxic language, harmful imagery, content moderation, and PII. Supports role-based access control and data governance. Ensures zero data retention for external model providers. Offers resources and best practices for AI Governance Reviews and establishing an internal AI governance office.	Stated as fully encrypted and completely secure .
Testing & Monitoring	Features a comprehensive METRO (Monitoring, Evaluations, Tracing, Reporting, Observability) framework. Supports automated testing , correctness evaluation using "LLM as a Judge", and tracking of accuracy, latency, token usage, and safety flags . Provides step-by-step execution timelines for tracing and a centralized dashboard for activity overview.	Offers transparency monitoring agent performance and consumption . (The description is less detailed compared to Oracle's METRO framework).
Pricing Model	Embedded AI features are included at no additional cost with existing Fusion subscriptions. OCI-hosted LLMs (Llama, Cohere) are no-charge with unlimited, unmetered use . Custom AI Agents have predictable monthly fees per user or per employee , avoiding consumption-based billing for predictability. Development costs are zero; subscription is only required when publishing to users. Includes a generous allocation of free OpenAI tokens (at least 100 million per month).	Promoted with the tagline: "All starting at zero cost." (No further details on potential tiered pricing or specific usage limits are provided in the source).

Unique Selling Points / Advantages	Deep native integration with Oracle Fusion Applications , allowing agents to understand business data and context effectively. Strong emphasis on enterprise-level task automation across various business functions (HR, Finance, SCM, Sales, Service). Comprehensive governance and compliance capabilities built-in. Robust METRO tools for detailed quality, performance, and cost optimization.	Focuses on democratizing access to the full potential of generative AI . Highlights its user-friendly drag-and-drop interface that requires no technical background. Emphasizes its "zero cost" starting point for all mentioned capabilities.
------------------------------------	--	---